

Q3. Sales Forecasting Dashboard

A retail company wants to analyze monthly sales data and predict future sales trends.

Explain

how:

A. Pandas is used for reading and cleaning sales data

B. SciPy and Scikit-Learn are used for analysis and prediction

C. Matplotlib and Seaborn are used to create business visualizations

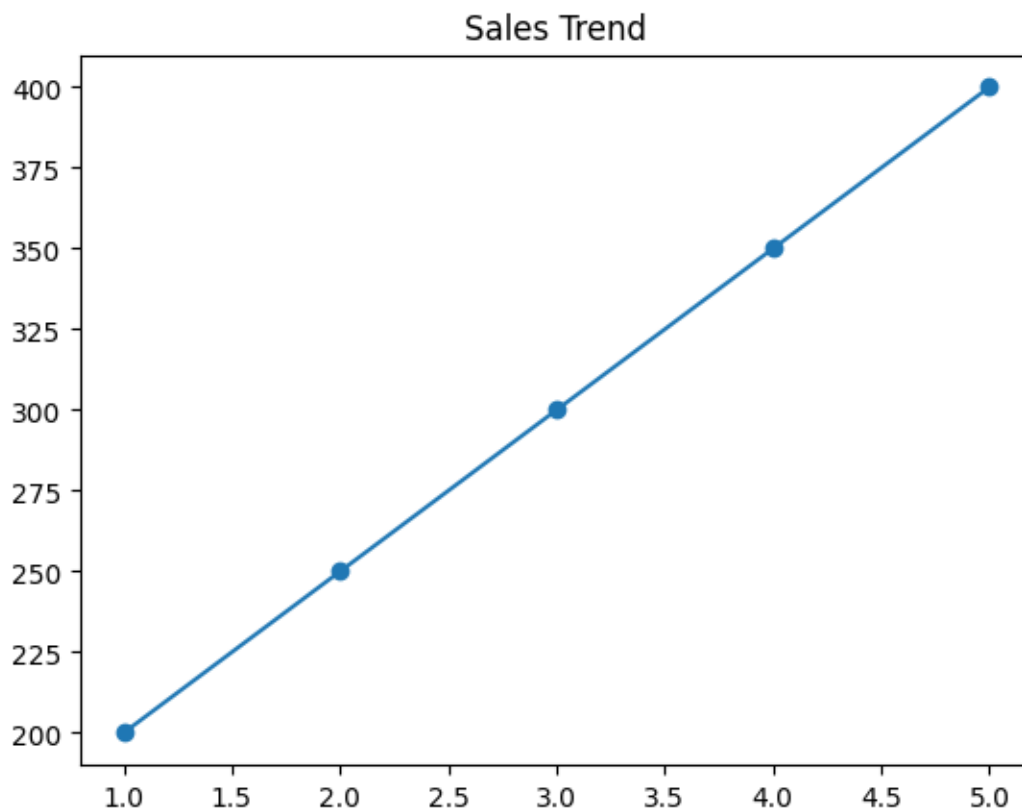
CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

data = {
    "Month": [1, 2, 3, 4, 5],
    "Sales": [200, 250, 300, 350, 400]
}

df = pd.DataFrame(data)
X = df[["Month"]]
y = df["Sales"]
model = LinearRegression()
model.fit(X, y)
future_months = np.array([[6], [7]])
predictions = model.predict(future_months)
print("Future Sales:", predictions)
plt.plot(df["Month"], df["Sales"], marker='o')
plt.title("Sales Trend")
```

```
plt.show()
```



PROBLEM:

The retail company wants to **analyze past sales data and predict future sales**. They need better decision-making for inventory, marketing, and revenue growth.

METHODOLOGY:

- 1. Data Collection**
Gather historical sales data from databases or files.
- 2. Data Preprocessing**
Clean the data using Pandas (handle missing values, remove errors).
- 3. Data Analysis**
Analyze trends, seasonality, and patterns using statistical methods.
- 4. Model Building**
Use machine learning models (like Linear Regression) for forecasting.
- 5. Evaluation**
Check model accuracy using performance metrics.

6. **Visualization**

Present results using charts.

SOLUTION:

The solution is to build a **Sales Forecasting Dashboard**.

- Use Pandas for data handling
- Use Scikit-Learn for prediction models
- Use visualization tools for insights
- Deploy as a dashboard for business users